WHEN TRADE STOPS AT SEA

Risk Management Challenges for Pakistan's Import & Export Industry Through the Strait of Hormuz

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20 mb/d
DAILY OIL TRANSIT

~25%
GLOBAL SEABORNE TRADE

2/3
LNG VIA HORMUZ

$50B
POTENTIAL ANNUAL LOSS

Executive Summary

PMBOK Risk Management Framework & Priority Actions

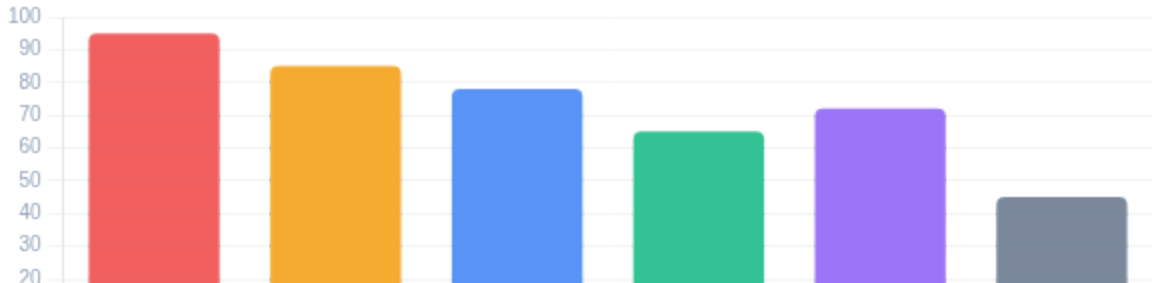
Risk Assessment Complete

PMBOK Risk Management Process

1. Plan Risk Management
2. Identify Risks
3. Qualitative Analysis
4. Quantitative Analysis

Top Priority Risks

LNG Supply Squeeze	Critical Dependency	95%
Insurance Shock	High Impact	85%
Freight Surcharge	Medium Impact	78%
Schedule Unreliability	Medium Impact	65%



$50B	31%	2/3
POTENTIAL LOSS	ENERGY IMPORTS	LNG VIA HORMUZ

Priority Actions for Risk Mitigation

Build Strategic Reserves — HIGH
Establish 30-60 day safety stock for critical inputs

Diversify Suppliers — HIGH
Reduce single-corridor dependence beyond Gulf

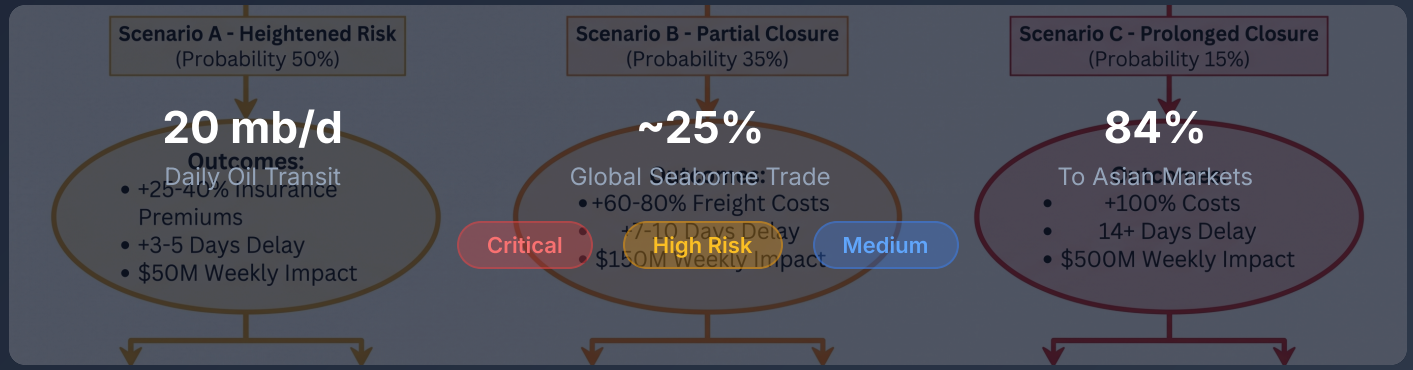
Port Coordination — MEDIUM
Integrate Karachi, Port Qasim, Gwadar operations

Global Importance of Strait of Hormuz

Critical chokepoint for global energy security and trade flows

Strategic Analysis

Strait of Hormuz - Global Trade Corridor



Oil Transit

20 mb/d

Daily oil flow through strait

↑ 25% of global trade

LNG Trade

~20%

Global LNG transit

⚠ 2/3 via Hormuz

Risk Distribution



Strategic Significance

The Strait of Hormuz is one of the world's most critical maritime chokepoints. In 2024-2025, oil flow averaged **20 million barrels per day**, representing ~25% of global petroleum liquids consumption. Limited bypass options

May 2026 | Global Trade Analysis | Maritime Security

View Analysis | Download

Pakistan's Dependency & Routes

Gulf Shipping Corridor Analysis - Maritime Trade Infrastructure

Maritime Analysis

Gulf-Pakistan Maritime Corridor

Strategic Route

Qatar/UAE → Hormuz → Pakistan

- Distance: ~1,200 nautical miles
- Transit time: 5-7 days
- Critical chokepoint: Hormuz

Key Metrics

- 2/3 LNG via Hormuz
- ~80% oil imports Gulf
- $12B+ annual trade

Karachi
Active

Port Qasim
Active

Gwadar
Developing

LNG Dependency
66.7%
+2.1%

Oil Import Exposure
~80%
+1.5%

Annual Trade Value
$12B+
+8.2%

95%
Route Utilization

7 days
Avg Transit

PMBOK Risk Identification & Register

Risk Breakdown Structure (RBS) & Detailed Risk Assessment

RBS Framework | Risk Register

Risk Breakdown Structure

- External/Geopolitical — 12
- Market/Commercial — 8
- Operational/Logistics — 15
- Technical — 6
- Legal/Regulatory — 4
- Environmental — 3

48 TOTAL RISKS

12 CRITICAL

Risk Register - Detailed Assessment

ID	Risk Name	Probability	Impact	Owner	Response Strategy
R001	LNG Supply Squeeze	95%	Critical	Energy Dept	Diversify suppliers, build 30-60 day buffers, FSRU deployment
R002	Insurance Premium Spike	85%	High	Finance	War-risk insurance pools, hedging strategies
R003	Freight Surcharge	78%	High	Logistics	Dynamic scheduling, alternative routes
R004	Schedule Unreliability	65%	Medium	Operations	Buffer time, contingency planning
R005	FX Volatility	72%	Medium	Treasury	FX hedging, forward contracts
R006	Container Imbalance	58%	Medium	Supply Chain	Inventory management, supplier coordination
R007	Port Damage	25%	High	Port Authority	Emergency protocols, insurance coverage

May 2026 | PMBOK Framework | 48 Risks Identified

Export Register | Filter

Qualitative Risk Analysis - Probability & Impact Matrix

5×5 Matrix Assessment for Pakistan Hormuz Trade Risks

Risk Assessment

Probability-Impact Matrix (5×5)

Impact →	Very Low	Low	Medium	High	Very High
Very High	1	2	3	4	5
High	6	7	8	9	10
Medium	11	12	13	14	15
Low	16	17	18	19	20
Very Low	21	22	23	24	25

Critical (90-100%) High (70-90%) Medium (40-70%) Low (0-40%)

Risk Details

LNG Supply Squeeze R001
Probability: 95% | Impact: Very High

Insurance Shock R002
Probability: 85% | Impact: High

Freight Surcharge R003
Probability: 78% | Impact: High

Schedule Unreliability R004
Probability: 65% | Impact: Medium

FX Volatility R005
Probability: 72% | Impact: Medium

Container Imbalance

8	3	5
TOTAL RISKS	CRITICAL	HIGH

May 2026 Risk Analysis 8 Active Risks

Export Matrix Filter

Stakeholder Risk Appetite Assessment

Likert Scale Analysis Across Risk Dimensions

Stakeholder Analysis

Risk Appetite by Stakeholder

Government

Policy & Regulation

Schedule	3.0
Cost	3.0
Safety	4.0
Legal	3.0

Moderate

Ports Authority

Operations & Logistics

Schedule	3.0
Cost	3.0
Safety	4.0
Legal	3.0

Moderate

Importers

Trade & Commerce

Schedule	2.0
Cost	2.0
Safety	3.0
Legal	2.0

Low

Risk Appetite Distribution



Low Risk (1-2) Medium Risk (3) High Risk (4-5)

1	2	3	4	5
Very Low	Low	Moderate	High	Very High

Quantitative Risk Analysis - Decision Tree

Disruption scenarios with cost impacts and mitigation strategies

Decision Analysis

Hormuz Disruption Event Analysis

May 2026 | Weekly Impact

Scenario A — 50%

Cost Impact	Delay
$50M	3-5 days

- Emergency stocks -$10M
- Insurance layers -$5M

Scenario B — 35%

Cost Impact	Delay
$150M	7-10 days

- Rerouting -$30M
- Alt suppliers -$20M

Scenario C — 15%

Cost Impact	Delay
$500M	14+ days

- Strategic reserves -$100M
- FSRU deployment -$50M

Expected Monetary Value (EMV) $152.5M

Scenario A (50% × $50M)	$25M
Scenario B (35% × $150M)	$52.5M
Scenario C (15% × $500M)	$75M
Total EMV	$152.5M

Risk Mitigation Impact



Run Simulation

Export

$152.5M	$7.9B	30-40%
BASELINE EMV	ANNUALIZED	RISK REDUCTION

Expected Monetary Value (EMV) Analysis

Quantitative Risk Assessment with Sensitivity Analysis & Mitigation Scenarios

EMV Calculation

Baseline EMV Calculation

(0.50 × $50M) + (0.35 × $150M) + (0.15 × $500M) = $152.5M

Scenario A (Heightened Risk)	$25M (50% × $50M)
Scenario B (Partial Closure)	$52.5M (35% × $150M)
! Scenario C (Prolonged Closure)	$75M (15% × $500M)

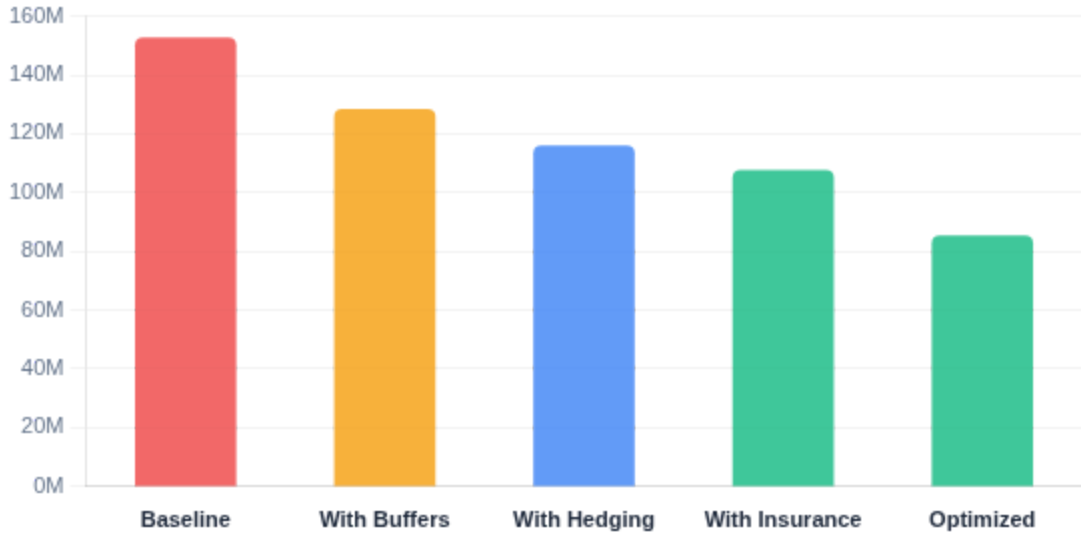
Total EMV (Weekly)	$152.5M
Annualized EMV	$7.9B

Sensitivity Analysis

Mitigation Impact Analysis

30-40%
Risk Reduction

$45M
Cost Savings



Mitigation Effectiveness
With buffers, hedging & insurance pooling

35%

May 2026 | Quantitative Analysis | EMV Calculation

Export Analysis

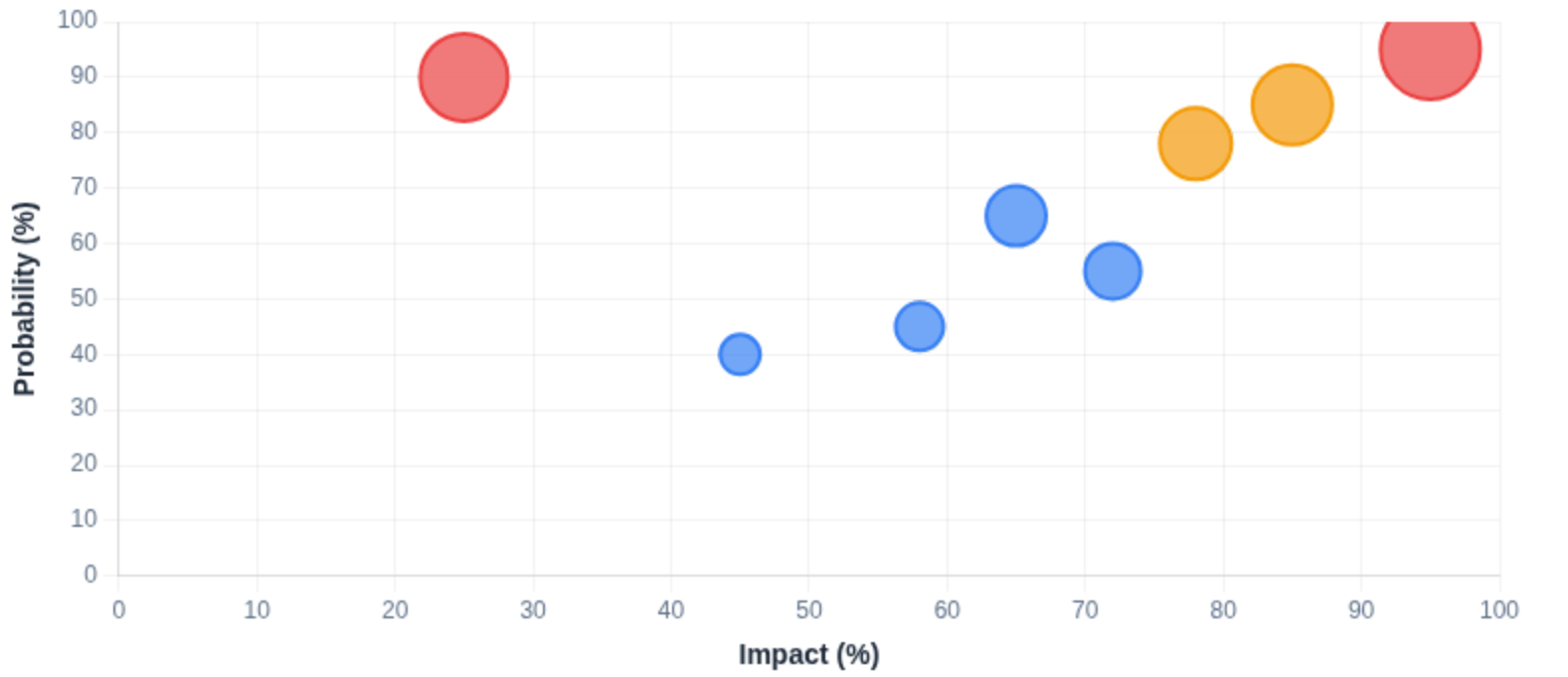
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Risk Prioritization - Bubble Chart Analysis

Probability vs Impact with Exposure/Value-at-Risk

Risk Analysis

Risk Bubble Chart



Critical (>90%) | High (70-90%) | Medium (40-70%) | Low (<40%)

Risk Summary

8

3	5
Critical	High

Priority Zones

Q1: High P, High I
Immediate Action

Q2: Low P, High I
Monitor Closely

Q3: High P, Low I
Regular Review

Q4: Low P, Low I
Accept Risk

R001	LNG Supply	0.95	Critical
R002	Insurance	0.85	High

May 2026 | Risk Analysis | 8 Active Risks

Export Chart | Filter

Risk Response Strategy - PMBOK Framework

Four response strategies with cost-to-mitigate analysis and ownership assignment

Response Planning

PMBOK Risk Response Strategies

Total: $152.5M EMV

AVOID — $25M

- Alternative routes via Red Sea
- Pipeline development
- Reduce dependency

Owners: Government, Ports

TRANSFER — $15M

- War-risk insurance pools
- FX hedging instruments
- Freight derivatives

Owners: Insurers, Importers

MITIGATE — $85M

- Diversify suppliers
- 30-60 day buffers
- FSRU deployment

Owners: Importers, Ports

ACCEPT — $27.5M

- Contingency reserves
- Emergency response fund
- Buffer capital

Owners: Government, Importers

Cost vs. Residual Risk

$152.5M

Avoidance Cost	$25M
Transfer Cost	$15M
Mitigation Cost	$85M
Acceptance Cost	$27.5M

Trigger Conditions

- Insurance premiums >25%
- Delays >7 days
- Route closure
- Freight costs >50%

May 2026 | Risk Analysis | 4 Stakeholders

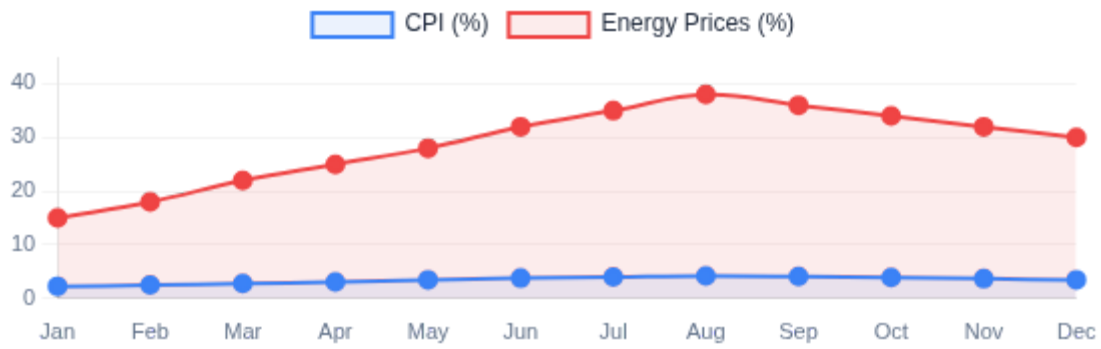
Run Analysis | Export

Macroeconomic Impact Analysis

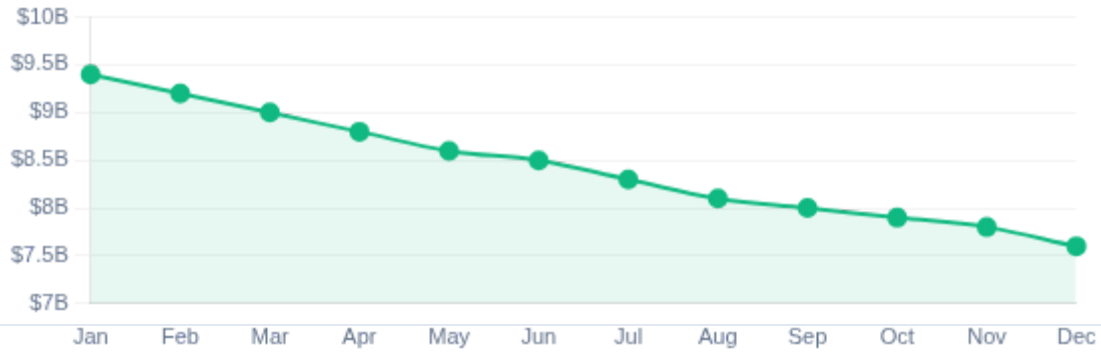
CPI inflation, FX reserves, GDP impact, and export decline under disruption scenarios

Economic Analysis

CPI vs Energy Prices



FX Reserves Trajectory



+4.2%	-8.5%	+$2.8B	-1.5%
CPI Impact	FX Reserves	CAD Impact	GDP Impact

Scenario Analysis

A B C

Industry Output	Export Decline
-25%	-5.2%
↓ -2.5% from baseline	↓ -1.2% from baseline

Energy Cost	Import Cost
+15%	+12%
↑ +3.5% from baseline	↑ +2.8% from baseline

Export Report Filter

May 2026 Macroeconomic Analysis 3 Scenarios

Implementation Roadmap

90-day immediate actions → 12-18 month strategic deployment

Timeline

Project Timeline & Milestones

35% Complete

Phase 1: Immediate Actions

Active

0-90 Days

- **Insurance Review** — Update marine insurance coverage — In Progress
- **Safety Stocks** — 30-60 days critical inputs — Done
- **FX Hedging** — Hedge energy imports — In Progress
- **Priority Berthing** — Emergency berthing windows — Pending

4	75%
Tasks	Progress

Owners: Government, Ports

Phase 2: Expansion

Pending

3-6 Months

- **Supplier Diversification** — Multi-source procurement — Pending
- **Swap Clauses** — Flexible volume agreements — Pending
- **Digital Slotting** — Automated scheduling — Pending
- **Alternative Routes** — Red Sea contingencies — Pending

4	0%
Tasks	Progress

Owners: Importers, SBP

Phase 3: Optimization

Pending

6-18 Months

- **Storage Expansion** — Gwadar/Port Qasim — Pending
- **FSRU Deployment** — Floating storage units — Pending
- **GCC Protocols** — Convoy security — Pending
- **Insurance Pool** — Credit guarantees — Pending

4	0%
Tasks	Progress

Owners: Government, Ports

Conclusion & Recommendations

Strategic roadmap for building trade resilience through the Strait of Hormuz

Action Plan

Key Recommendations

1 **Strategic Reserves & Buffers**

Expand LNG storage, refined product reserves, and FSRU options for energy security. Build 30-60 day safety stock for critical inputs.

30-60 days stock | FSRU deployment

2 **Diversify Suppliers**

Reduce single-corridor dependence by exploring alternative suppliers beyond Gulf. Multi-source procurement from Qatar, UAE, and alternative suppliers.

Multi-source | Swap clauses

3 **Port Coordination**

Integrate Karachi, Port Qasim, Gwadar operations with emergency protocols. Prioritize berthing for energy and critical cargo.

Emergency berthing | Digital slotting

4 **Financial Risk Tools**

Expand FX planning, freight risk management, and insurance coverage. Implement FX hedging strategies for energy imports.

FX hedging | Insurance layers

5 **Public-Private Response Cell**

Create public-private maritime disruption response coordination unit with real-time monitoring and coordinated response.

Success Metrics

Target KPIs

30% Cost Reduction ↑ Target	15% Risk Mitigation ↑ Target
95% Supply Uptime ✓ Target	3 days Response Time ↓ From 7 days

Implementation Status

4/5
Completed

90 days
Timeline

Export Plan

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